Rejection Sampling for Medical MCQ Generation and Fine-Tuning TinyLlaMA for Layperson QA

Chelsea Ramos
University of Texas at Austin
chelseanbr@utexas.edu

PowerPoint Slides: PowerPoint Slides
Video Presentation: Video Presentation
Code Repository: github.com/chelseanbr/aih-project
Data Source: huggingface.co/datasets/lavita/MedQuAD

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Chelsea Ramos
University of Texas at Austin
Austin, Texas, USA
chelseanbr@utexas.edu

1 Introduction

Medical question answering is a challenging task due to the complexity of medical language and the need for accurate information. In this project, I focus on generating multiple-choice questions (MCQs) from the MedQuAD layperson Question Answering dataset via Rejection Sampling. The goal is to create a set of MCQs that can be used to assess layperson medical knowledge. Additionally, I fine-tune the TinyLLaMA model to improve its performance on layperson questions.

2 Related Work

Discuss previous work here. Citation [1].

3 Methodology

Explain what you did.

4 Results

Show results here.

5 Conclusion

Summarize and future directions.

References

[1] Asma Ben Abacha and Dina Demner-Fushman. 2019. A Question-Entailment Approach to Question Answering. *BMC Bioinformatics* 20, 1 (2019). https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-019-3119-4 Article number 511:1–511:23.

[2] Zheng Yuan, Hongyi Yuan, Chengpeng Li, Guanting Dong, Keming Lu, Chuanqi Tan, Chang Zhou, and Jingren Zhou. 2023. Scaling Relationship on Learning Mathematical Reasoning with Large Language Models. *arXiv preprint arXiv:2308.01825* (2023). https://arxiv.org/abs/2308.01825

[3] Jiean Zhu. 2024. Cura-LLaMA: Evaluating Open-Source Large Language Model's Question Answering Capability on Medical Domain. *Applied and Computational Engineering* 90 (2024), 52–60. doi:10.54254/2755-2721/90/20241725

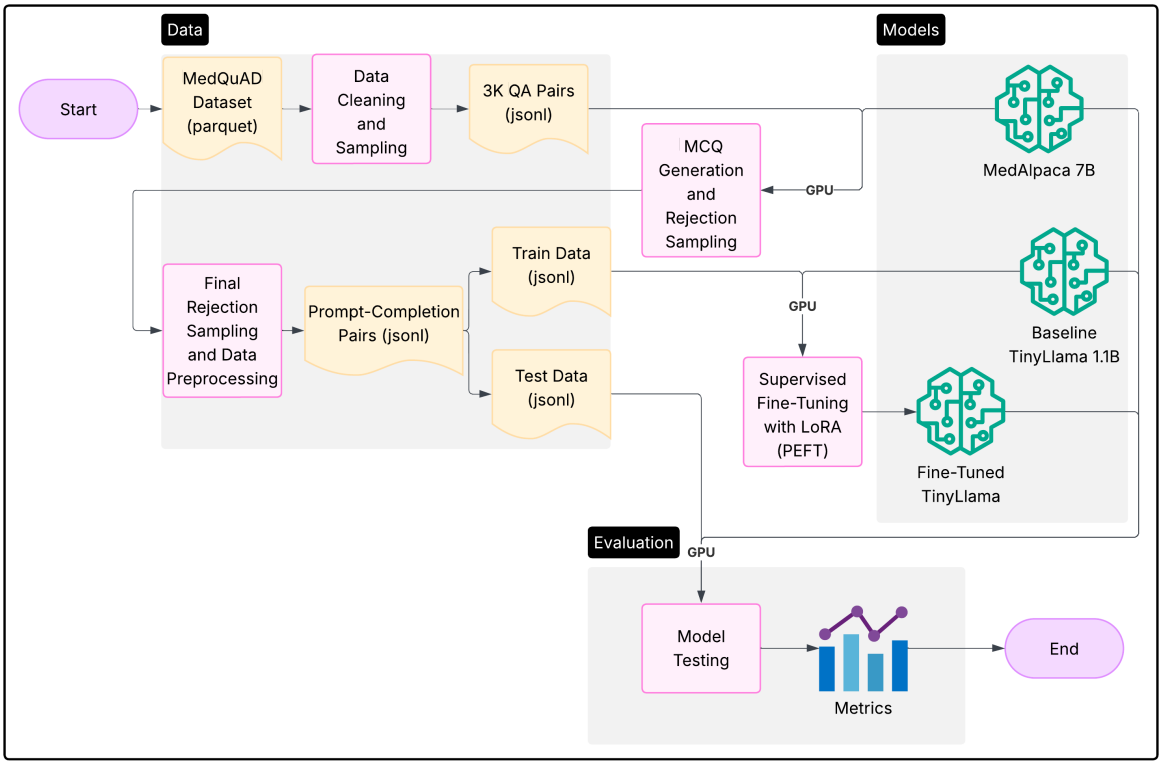

Figure 1: Workflow for MCQ Generation and Model Fine-Tuning